Agile Project Management

Chapter 4: Sprint Execution

Chapter Objectives

In this chapter, we will:

- Conduct a sprint planning session
- Follow the process as the Scrum Master
- Review some of the engineering and mechanical practices of agile
- Bring the sprint to a graceful close as the product owner

Chapter Concepts

- **Sprint Planning Session**
- Sprint Process (Scrum Master)
- Mechanics and Engineering Practices
- End of Sprint Review (Product Owner)
- Chapter Summary

Sprint Planning Meeting

Typically with the product owner

Part 1 – What

Carve items from the product backlog

Establish a sprint goal

Identify the sprint backlog

Typically just the development team

Part 2 – How

Define a plan to create the "done" increment

Assign the tasks to the team

The work to be performed in the sprint is planned at the Sprint Planning Meeting. This plan is created by the collaborative work of the entire Scrum Team.

Story Points

- Arbitrary measure to gauge the relative difficulty of completing any item on the product backlog
- Process:
  - Pick an item on the product backlog that the team has a good understanding of
    - Preferably a small item
  - Assign a number to it, usually 3, 4, or 5
  - This is its story point value
  - Weigh all other items on the product backlog in relation to the difficulty of doing the selected item
    - If harder or more complex, the story point number is higher
    - If easier, the story point number is less

Planning Poker

1. Each participant gets a deck of estimation cards representing a sequence of numbers. The most popular sequences involve doubling each card (0, ½, 1, 2, 4, 8, 16, etc.) or the Fibonacci sequence (1, 2, 3, 5, 8, 13, 21, etc.). The Fibonacci sequence is generally more popular.
2. The moderator presents one user story at a time to the team.
3. The product owner (or equivalent role) answers any questions the team might have about the story.
4. Each participant privately selects a card representing his or her estimate of the "size" for the user story. Usually size represents a value taking into account time, risk, complexity, and any other relevant factors.

Planning Poker (continued)

5. When everybody is ready with an estimate, all cards are presented simultaneously.
6. If there is consensus on a particular number, then the size is recorded and the team moves to the next story.
7. In the (very likely) event that the estimates differ, the high and low estimators defend their estimates to the rest of the team.
8. The group briefly debates the arguments.
9. A new round of estimation is made (steps 4 and 5 above).
10. Continue until consensus has been reached and the moderator records the estimate.
11. Repeat for all stories.

Exercise 4.1: User Story Estimations (Optional)

GROUP EXERCISE

- Using the assigned product backlog:
  - Begin at the top estimating the backlog items (BLI)
  - If you find a backlog item estimate greater than your sprint length, split the backlog item into smaller stories
  - Estimate only stories, not epics
  - Place the estimate next to the BLI on the flip chart
  - When you have “filled” your sprint, estimate two more backlog items and stop

Velocity

- Velocity is the speed at which the team can work
  - Usually based on experience
  - Adjusted over each sprint
  - Weighed against capacity
- Velocity is based on:
  - People
  - Process
  - Product
    - Selecting the appropriate product size
  - Technology
    - Using the most effective tools
    - Balancing risk with new technology
- Takes about three sprints to be able to gain a relatively predictable measure of velocity

Velocity is not a measure of productivity

Sprint Plan Verification

- Do we all understand everything we are doing in the sprint?
- Are we asking people to do too many things in this sprint?
- Are there too many topics to focus on?
- Is there an overload on one or two people with a certain skill set?
- Are we so heavily focused in one area that some people on the team will have “no” work?
- Are all the “outside” things lined up, such as people you need to collaborate with, their managers, alignment on your approach or philosophy, etc.

Chapter Concepts

Sprint Planning Session

Sprint Process (Scrum Master)

Mechanics and Engineering Practices

End of Sprint Review (Product Owner)

Chapter Summary

Sprint Meetings

- Scrum team meets daily (Daily Scrum)
  - Progress reviewed
  - Impediments identified
  - Provides feedback on progress
- At the end of Sprint, Scrum team meets with management (sprint review meeting, retrospective)
  - Inspect product increment
  - Product backlog rearranged based on Sprint results

Daily Scrum or Stand-Up Meetings

- During the Daily Scrum, each team member typically answers three questions
  - What did I complete yesterday?
  - What do I plan to complete today?
  - Do I see any impediment for the team or myself?
- The Daily Scrum
  - Starts precisely on time even if some development team members are missing
  - Should happen at the same time and place every day
  - Is limited to 15 minutes
- Well organized Daily Scrums:
  - Improve communication
  - Enable progress to be accurately tracked
  - Promote quick decision making
  - Eliminate other meetings
  - Identify and remove impediments

What Is NOT a Daily Scrum Meeting?

- NOT a venue to discuss technical details, argue, or design a solution
  - Should be taken offline and separate meetings are to be set up
- NOT a project status update
  - Primary goal of Daily Scrum is to plan future work, not to discuss work that has been done
- It is for the TEAM and NOT for Scrum Master
  - Scrum Master sets up the meeting and facilitates, but team does NOT report to him
- NOT a planning meeting
  - Any big changes that can affect achieving sprint goal should be discussed in a separate meeting

Follow-Up Meetings

- Team members can request follow-up meetings during Daily Scrum
  - Referred to as working sessions
- Number of reasons for these sessions including:
  - One team member knows a solution to an impediment
  - A team member wants to share information
  - A design decision needs to be made
  - Technology choice must be made
- Any team member is welcome to attend
  - Held immediately after Daily Scrum where session was requested
- Helps maintain Daily Scrum as focused and short

Scrum Master's Role in Daily Scrum

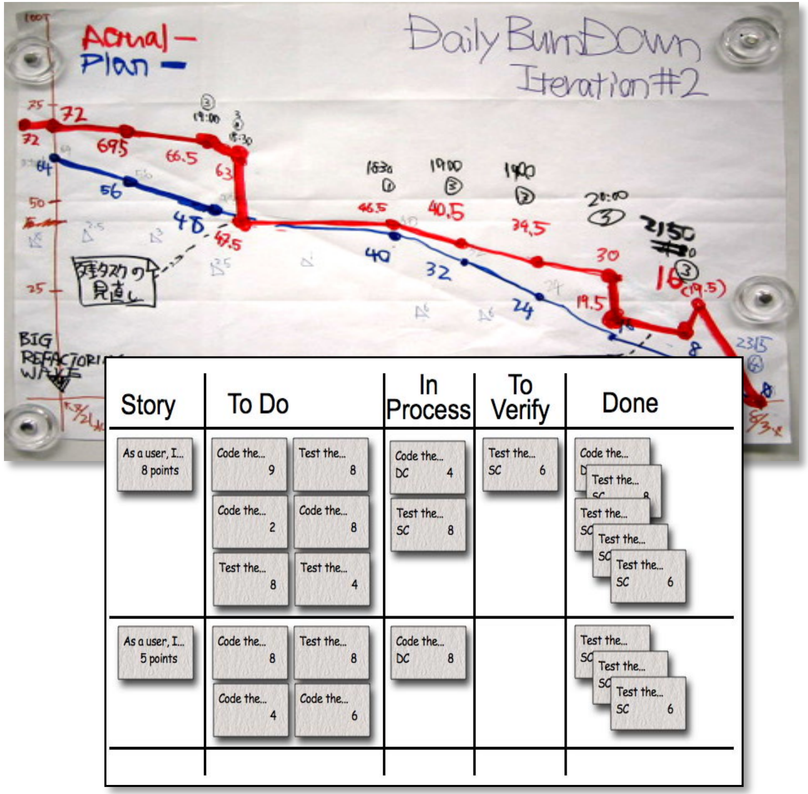
- Responsible for conducting Daily Scrum
  - Time and location
- Establishing layout of the meeting room
  - Ideally, table with just enough chairs for team members
  - Whiteboards for recording impediments, progress, etc.
  - Door to be closed to prevent interruptions!
- Meeting enables Scrum Master to sense how team is progressing
  - Individuals becoming swamped
  - Members lost interest
  - Somebody not working efficiently
  - Team chemistry

Identifying Impediments

- Scrum Master is responsible for removing impediments
  - Should be recorded on whiteboard
- Common impediments include:
  - Unsure of how to best use technology
  - Design decision needs to be made
  - Asked by management to undertake a different task
  - Infrastructure (server, network, workstation) slow
- Scrum Master must do all that is possible to remove impediments
  - If not removed, should be reported next day

Burn Down Charts and Backlogs

- Backlogs contain the stories and tasks that the Scrum Team must do
  - In this sprint
  - For the project
- Burn down charts show the progress toward the project goals
- Backlogs contain testing and QC tasks, as well as development tasks



Chapter Concepts

Sprint Planning Session

Sprint Process (Scrum Master)

Mechanics and Engineering Practices

End of Sprint Review (Product Owner)

Chapter Summary

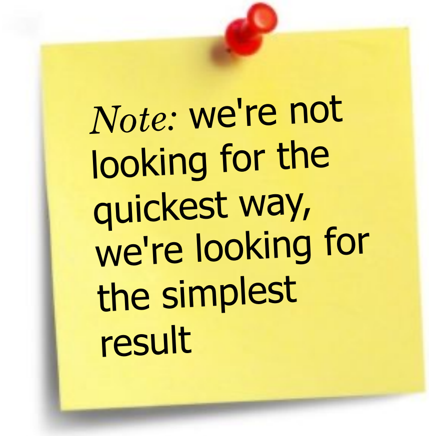
Technical Debt

- Doing things the quick and dirty way sets us up with a technical debt
  - Similar to a financial debt
- Technical debt incurs interest payments
  - The extra effort that we have to do in future development because of the quick and dirty design choice
- We can choose to continue paying the interest or we can pay down the principal by refactoring the quick and dirty design into the better design
  - Although it costs to pay down the principal, we gain by reduced interest payments in the future
- Sometimes it may be sensible to do the quick and dirty approach
  - Developers may purposefully incur technical debt to hit an important deadline
- All too common problem: development organizations let their debt get out of control and spend most of their future development effort paying crippling interest payments

Source: Ward Cunningham

Do the Simplest Thing that Could Possibly Work

- Consider all solutions that could work and select the simplest
- Make the code pass the unit/acceptance tests for new features
- Refactor the system to be the simplest possible code including all the features it now has
  - Break the thing into pieces
  - Modify one of the small pieces
- Applied to each increment
- Differentiate between “could possibly work” and “will work”



Pair Programming

- All production code is written with two people looking at one machine with one keyboard and one mouse [Ron Jeffries]
- It’s actually good to pair a designer with a coder or a user interface expert with a coder. They can learn a lot about each other’s area of expertise and partner in getting the whole job done.*
- In practice, coding is:
  - Less susceptible to interruptions
  - Much faster discovery of defects

- Each developer plays a different role and trades roles frequently (hourly basis)
  - Driver has keyboard and mouse and writes code
  - Navigator reviews what driver is writing
- Pairs switch off during given sprint or iteration so that every programmer pairs with everyone else at some point

*Laurie Williams, *Pair Programming Illuminated*, Addison-Wesley, 2003

Discussion: Will It Work?

WRITTEN EXERCISE

- Can you see pair programming working in your shop?

Positive Aspects of Pair Programming	Concerns About Pair Programming

Refactoring

- **Refactoring** is a term used to describe techniques that reduce the short-term pain of redesigning
  - A change made to the internal structure of the software to make it easier to understand and cheaper to modify without changing its observable behavior*
- Functionality of your program or design is not changed
- Internal structure to make it easier to understand and work with
  - Examples:
    - Breaking programs into smaller modules to reduce complexity
    - Removing unused or unreachable code
    - Following the rule of YAGNI (You Aren't Going To Need It)
    - Creating the simplest solution that will work

> Containing this software entropy is the mission of refactoring.
>
> —Peter Coffee

*Source: Martin Fowler, *Refactoring: Improving the Design of Existing Code*, Addison-Wesley, 1999

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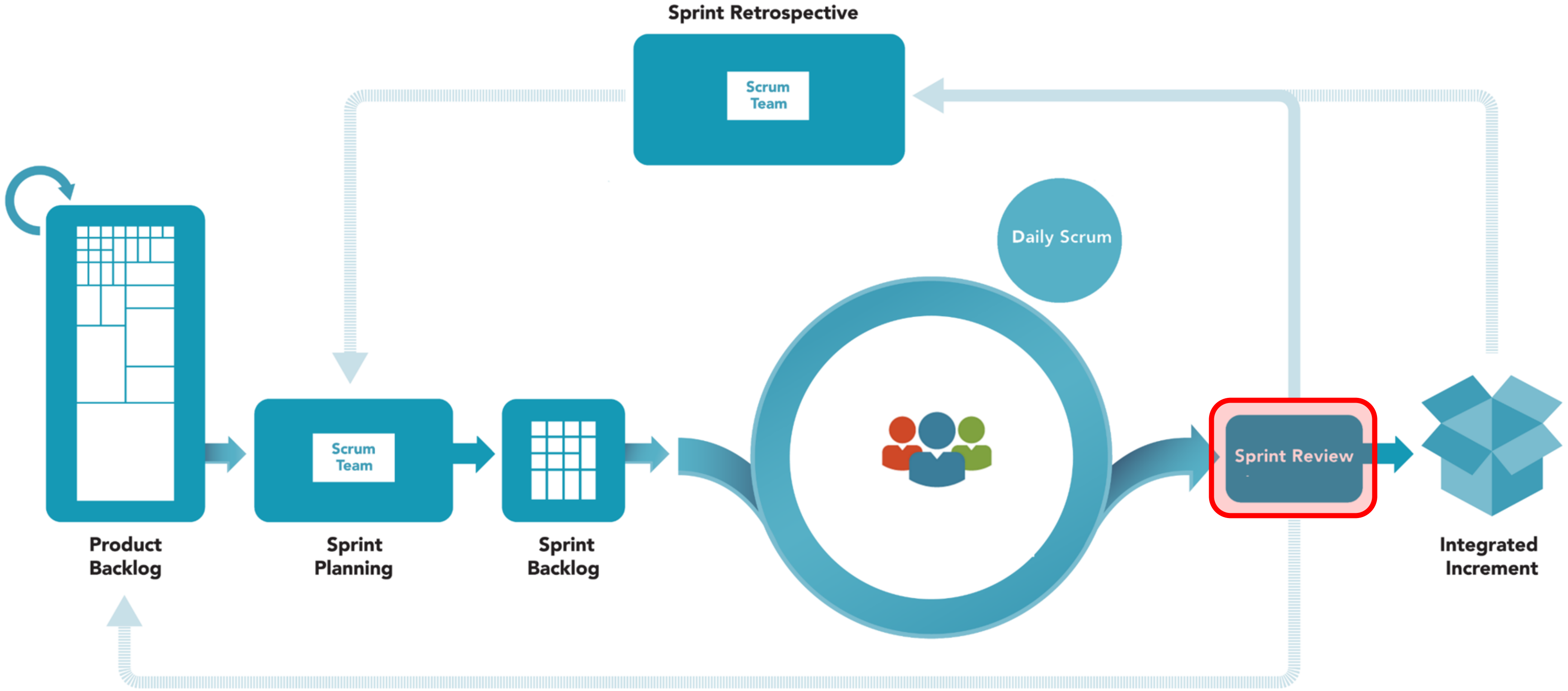

Sprint Retrospective
Scrum Team
Daily Scrum
Scrum Team
Product Backlog
Sprint Planning
Sprint Backlog
Sprint Review
Integrated Increment

Sprint Review Meeting

- Held at the end of the sprint to inspect the increment and adapt the product backlog if needed
- The Scrum Team and stakeholders collaborate about what was done in the sprint
- The Sprint Review includes the following elements:
  - The product owner identifies what has been “done” and what has not been “done”
  - The development team discusses what went well during the sprint, what problems it ran into, and how those problems were solved
  - The development team demonstrates the work that it has “done” and answers questions about the increment
  - The product owner discusses the product backlog as it stands. He or she projects likely completion dates based on progress to date.
  - The entire group collaborates on what to do next, so that the Sprint Review provides valuable input to subsequent sprint planning meetings

Exercise 4.2: Update the Product Backlog

- The instructor will give you an update on your backlog based on a sprint review session
  - Add new items based on product review
  - Change items and/or priorities
- As a business analyst working with the product owner:
  - Review the user stories you have created for the case study
  - Update the product backlog with the user stories as appropriate
  - Put a business value on each new item added to the product backlog
  - Re-order the product backlog in business value order

Sprint Retrospective

- A meeting to review and discuss things that did or did not work $\mathit{within\ team}$ during the last sprint
  - The focus is on HOW things where done and find areas of improvement
  - Meeting is 'inward' looking
    - This is the time and the place for team to praise each other or bring up some grudges in a safe, non-judgmental environment
- Usually the last thing done in a sprint
  - Many teams will do it immediately after the sprint review
  - Normally takes about an hour, though much take longer
- The entire Scrum team normally participates
  - This does NOT include external stakeholders or anyone else outside of Scrum team

Exercise 4.3: Retrospective (Optional)

GROUP EXERCISE

- Perform a project retrospective
  1. Review what was done in this project
  2. Determine what should be done in the next release
  3. Record lessons learned

> Regardless of what we discover, we understand and truly believe that everyone did the best job they could, given what they knew at the time, their skills and abilities, the resources available, and the situation at hand

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